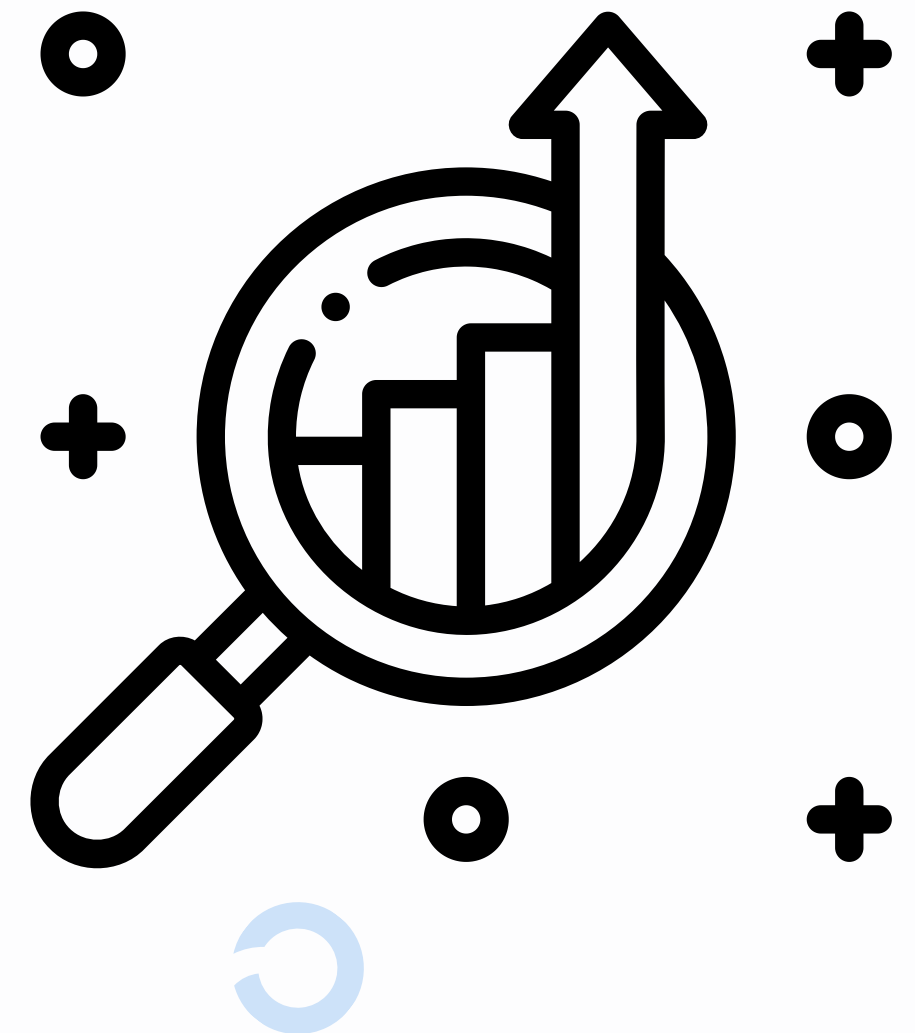
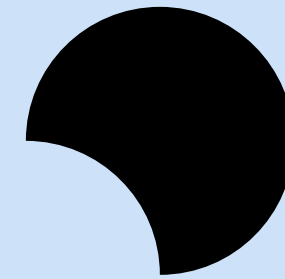


MARKETING CAMPAIGN

GROUP ZOAS
(ZUYING, AYANA, SOTA)



CAMPAIGN GOALS

kaggle

- **Advice and Suggestions**
- **Maintain current consumers**
- **Create more value**



OVERVIEW

- 1. Marketing and Business Background**
- 2. Problem Statement**
- 3. Descriptive Analytics**
- 4. Diagnostic Analytics**
- 5. Predictive Analytics**
- 6. Evaluation**
- 7. Conclusion**



STATEMENT OF THE PROBLEMS

What Is Marketing?

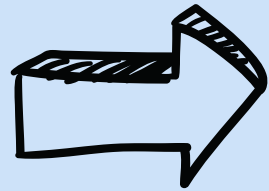
How does it connect to consumers and bring more value?

- Marketing refers to activities a company undertakes to promote the buying or selling of a product or service. Marketing includes advertising, selling, and delivering products to consumers or other businesses. (1)
- Marketing for business is a highly valuable tool as it helps to create brand awareness, drive profit and growth, acquire and retain customers, and enhance engagement.(2)
- The importance of understanding the relationship between marketing value and customers:
“The marketer has to understand what is of greatest value to the target customer, and then use that information to develop a total offering that creates value...” Marketing exists to help organizations understand, reach, and deliver value to their customers. For this reason, the customer is considered the cornerstone of marketing.(3)

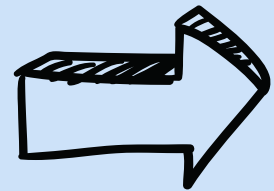
QUESTION:

What type of people is most likely to respond to marketing campaign when buying products?

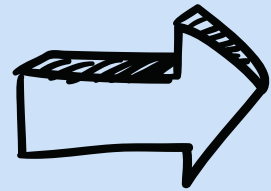
- Business Goals: To answer the question and make a new marketing strategy to maintain the current customers on the basis of data analytics.



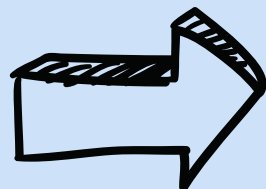
Find the characters of buyers by using dataset.



Via Pearson's Chi-squared test and Welch Two Sample T-test, what kind of people would most likely to buy products?



Via the Logistic Regression Model, to whether the future customers will respond to the campaign or not?



Is it there any relationship between people's characters and responses? How frequently do customers buy products in different characters? Which group tends to buy more?



DATASET

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC			
1	ID	Year	Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Dt_Out	Client_Owner	Recency	MntWtWnes	MntFriTues	MntMeatProducts	MntFishProducts	MntSweetProds	MntGoldProds	NumDealsPurchases	NumWebPurchases	NumCatalogPurchases	NumStorePurchases	NumWebVisitsMonth	AcceptedCmp3	AcceptedCmp4	AcceptedCmp5	AcceptedCmp1	AcceptedCmp2	Complain	Z_CostContact	Z_Revenue	Response	
2	5524	1957	Graduation	Single	58138	0	0	2012-09-04	58	635	88	546	172	88	88	3	8	10	4	2	0	0	0	0	0	0	0	0	0	0	0	
3	2174	1954	Graduation	Single	46344	1	1	2014-03-08	38	11	1	6	2	1	6	2	1	1	2	1	2	5	0	0	0	0	0	0	0	0	0	0
4	4141	1965	Graduation	Together	71613	0	0	2013-08-21	26	426	49	127	111	21	42	1	8	2	10	4	10	4	0	0	0	0	0	0	0	0	0	0
5	6182	1984	Graduation	Together	26646	1	1	2014-02-10	26	11	4	20	3	5	2	2	2	2	2	0	4	4	0	0	0	0	0	0	0	0	0	0
6	5324	1981	PhD	Married	58293	1	1	2014-01-19	94	173	3	173	5	118	46	173	27	15	3	5	3	0	0	0	0	0	0	0	0	0	0	0
7	7446	1967	Master	Together	62513	0	0	2013-09-09	16	520	42	14	98	2	42	14	98	2	7	10	3	7	0	0	0	0	0	0	0	0	0	0
8	965	1971	Graduation	Divorced	55635	0	1	2012-11-13	34	235	65	164	50	49	27	4	7	4	3	4	10	3	0	0	0	0	0	0	0	0	0	0
9	6177	1985	PhD	Married	33454	1	1	2013-05-08	32	76	10	56	3	1	23	2	4	56	0	0	4	4	0	0	0	0	0	0	0	0	0	0
10	4855	1974	PhD	Together	30351	1	0	2013-06-06	19	14	0	24	3	3	2	1	3	0	0	2	1	2	0	0	0	0	0	0	0	0	0	0
11	5899	1950	PhD	Together	5648	1	1	2014-03-13	68	28	1	6	1	1	13	0	1	0	0	0	0	20	1	0	0	0	0	0	0	0	0	0
12	1994	1983	Graduation	Married		1	0	2013-11-15	11	5	5	6	0	2	1	1	1	0	0	2	7	0	0	0	0	0	0	0	0	0	0	0
13	387	1976	Basic	Married	7500	0	0	2012-11-13	59	6	16	11	11	1	16	1	2	1	3	8	0	0	0	0	0	0	0	0	0	0	0	0
14	2125	1959	Graduation	Divorced	63033	0	0	2013-11-15	82	194	61	480	225	112	30	1	3	4	8	2	0	0	0	0	0	0	0	0	0	0	0	0
15	8180	1952	Master	Divorced	59354	1	1	2013-11-15	53	233	3	53	3	5	14	3	6	1	5	6	0	0	0	0	0	0	0	0	0	0	0	0
16	2569	1987	Graduation	Married	17323	0	0	2012-10-10	38	3	14	17	6	1	5	1	1	0	3	8	0	0	0	0	0	0	0	0	0	0	0	0
17	2114	1946	PhD	Single	82800	0	0	2012-11-24	23	1006	22	115	59	68	45	1	7	6	12	3	0	0	0	1	1	0	0	0	0	0	0	0
18	9736	1980	Graduation	Married	41850	1	1	2012-12-24	51	53	5	19	2	13	4	3	3	0	3	8	0	0	0	0	0	0	0	0	0	0	0	0
19	4939	1946	Graduation	Together	37760	0	0	2012-08-31	20	84	5	38	150	12	28	2	4	38	1	6	7	0	0	0	0	0	0	0	0	0	0	0
20	6565	1949	Master	Married	76995	0	1	2013-03-28	91	1012	80	498	0	16	176	2	11	4	9	5	0	0	0	0	1	0	0	0	0	0	0	0
21	2278	1985	2n Cycle	Single	33812	1	0	2012-11-03	86	4	17	19	30	24	39	2	2	1	3	6	0	0	0	0	0	0	0	0	0	0	0	0
22	9360	1982	Graduation	Married	37040	0	0	2012-08-08	41	86	2	73	69	38	48	1	4	2	5	8	0	0	0	0	0	0	0	0	0	0	0	0
23	5376	1979	Graduation	Married	2447	1	0	2013-01-06	42	1	1	1	1	1725	1	3	15	0	28	0	1	0	0	0	0	0	0	0	0	0	0	0
24	1993	1949	PhD	Married	58607	0	23	2012-12-23	63	86	0	86	3	0	86	0	3	6	7	1	0	0	0	0	0	0	0	0	0	0	0	0
25	4047	1954	PhD	Married	65324	0	1	2014-01-11	0	384	0	0	0	32	5	0	6	2	2	0	0	0	0	0	0	0	0	0	0	0	0	0
26	1409	1951	Graduation	Together	40689	0	1	2013-03-18	69	270	3	27	39	6	99	7	7	2	5	1	0	0	0	0	0	0	0	0	0	0	0	0
27	7892	1969	Graduation	Single	18589	0	0	2013-01-02	89	6	4	25	15	12	13	2	2	1	3	7	0	0	0	0	0	0	0	0	0	0	0	0
28	2404	1976	Graduation	Married	53359	1	1	2013-05-27	4	173	4	30	3	6	41	4	5	1	4	7	0	0	0	0	0	0	0	0	0	0	0	0
29	5255	1986	Graduation	Single		1	0	2013-02-20	19	5	1	3	3	263	362	0	27	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
30	9422	1989	Graduation	Married	38360	1	0	2013-05-31	26	36	2	42	20	21	10	2	2	1	4	3	0	0	0	0	0	0	0	0	0	0	0	0
31	1966	1965	PhD	Married	84618	0	0	2013-11-22	96	684	100	801	21	66	0	1	6	9	10	2	0	0	0	1	0	0	0	0	0	0	0	0
32	6864	1989	Master	Divorced	10979	0	0	2014-05-22	34	8	4	10	2	2	4	2	3	0	3	5	0	0	0	0	0	0	0	0	0	0	0	0
33	3033	1963	Master	Together	38620	0	0	2013-05-11	56	112	17	44	34	22	89	1	2	5	3	3	0	0	0	0	0	0	0	0	0	0	0	0
34	5710	1970	Graduation	Together	40548	0	1	2012-10-10	31	110	0	5	2	0	3	2	2	1	4	5	0	0	1	0	0	0	0	0	0	0	0	0
35	7373	1952	PhD	Divorced	46610	0	2	2012-10-29	8	96	12	96	33	22	43	6	4	1	6	6	0	0	0	0	0	0	0	0	0	0	0	0
36	8755	1946	Master	Married	68657	0	0	2013-02-20	4	482	34	471	119	68	22	1	3	5	9	7	0	0	0	0	0	0	0	0	0	0	0	0
37	10738	1951	Master	Single	49389	1	1	2013-08-29	55	40	0	19	2	1	3	1	2	0	3	7	0	0	0	0	0	0	0	0	0	0	0	0
38	4339	1970	PhD	Married	67353	0	1	2013-12-31	37	702	17	151	0	8	35	5	5	6	12	2	0	0	0	0	0	0	0	0	0	0	0	0
39	10755	1973	2n Cycle	Married	23718	1	0	2013-09-02	76	6	3	14	15	7	36	3	3	1	2	7	0	0	0	0	0	0	0	0	0	0	0	0
40	8595	1973	Graduation	Widow	42429	0	1	2014-02-11	99	55	0	1	6	2	4	1	4	1	3	5	0	0	0	0	0	0	0	0	0	0	0	0
41	2968	1943	PhD	Divorced	48948	0	0	2013-02-01	53	437	4	206	160	7	437	5	437	7	10	5	0	0	1	0	0	0	0	0	0	0	0	0
42	8601	1980	Graduation	Married	80011	0	1	2013-04-29	3	421	76	536	2	178	102	2	8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
43	503	1985	Master	Married	20559	1	0	2013-03-12	88	13	1	29	3	0	7	2	2	0	3	1	0	0	0	0	0	0	0	0	0	0	0	0
44	8430	1957	Graduation	Together	21994	0	1	2012-12-24	4	9	0	6	3	1	3	1	0	0	3	5	0	0	0	0	0	0	0	0	0	0	0	0
45	7281	1959	PhD	Single		0	0	2013-11-05	80	81	11	50	3	2	39	1	1	3	4	2	0	0	0	0	0	0	0	0	0	0	0	0
46	2139	1975	Master	Married	7500	1	0	2013-10-02	19	3	1	10	3	2	12	3	2	0	3	5	0	0	0	0	0	0	0	0	0	0	0	0
47	1371	1976	Graduation	Single	79941	0	0	2014-06-28	72	123	164	266	227	30	174	1	2	4	9	1	0	0	0	0	0	0	0	0	0	0	0	0
48	9909	1996	2n Cycle	Married	7500	0	0	2012-11-09	24	3	18	14	15	22	50	3	3	1	3	0	0	0	0	0	0	0	0	0	0	0	0	0
49	7286	1968	Graduation	Together	41728	1	0	2013-05-24	92	13	6	15	3	5	13	3	2	0	2	10	0	0	0	0	0	0	0	0	0	0	0	0
50	7244	1951	Graduation	Single		2	1	2014-01-01	96	48	5	48	6	10	7	3	2	1	4	6	0	0	0	0	0	0	0	0	0	0	0	0
51	6566	1954	PhD	Married	72550	1	1	2012-11-08	39	826	50	317	50	38	38	9	5	2	12	8	0	0	0	0	0	0	0	0	0	0	0	0
52	8614	1957	Graduation	Widow	65486	0	1	2014-05-12	29	245	19	125	37	67	14	1	4	2	10	2	0	0	0	0	0	0	0	0	0	0	0	0
53	4114	1964	Master	Married	79143	0	0	2012-08-11	2	650	37	780	27	167	32	2	780	32	13	3	0	0	0	0	0	0	0	0	0	0	0	0
54	1331	1977	Graduation	Single	35790																											

Target & Definition(0,1)

Meat, Wine, Fish products...

Education, Marital Status, Kidhome...



What methods are we going to use?

DATA ANALYSIS

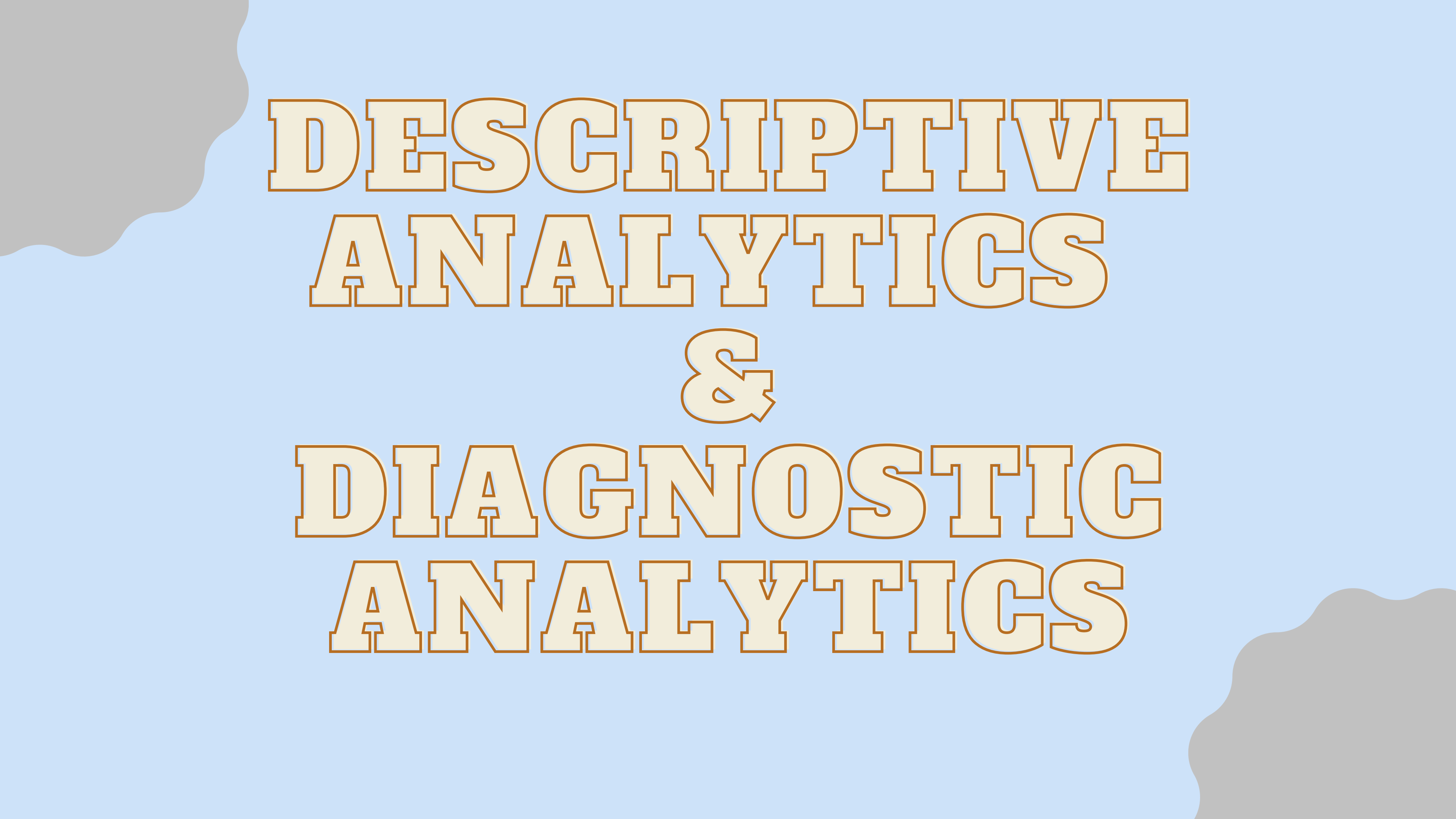
THE CRISP-DM METHODOLOGY

1. Data Understanding: Historical data from the company and use Chi-square test/T-test to do our analysis.

2. Data Preparation: We deleted some missing data and complaint factors that do not matter to our business problems.

3. Data Modelling: We build Logistic Regression model with feature engineering

4. Testing and Evaluation:
Evaluated by PR-AUC with Cross Validation.



DESCRIPTIVE ANALYTICS & DIAGNOSTIC ANALYTICS

Pearson's Chi-squared test

No Buy
Buy

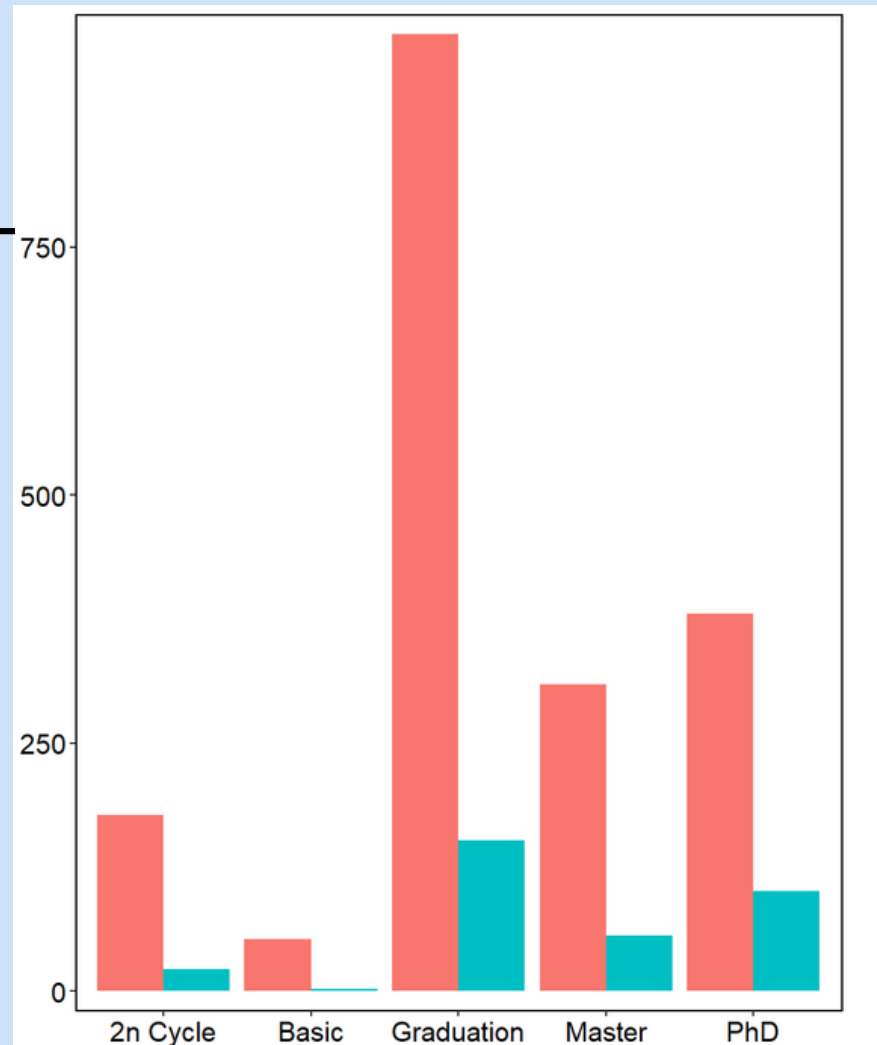
P-value = 1.18×10^{-4}

P-value = 2.94×10^{-9}

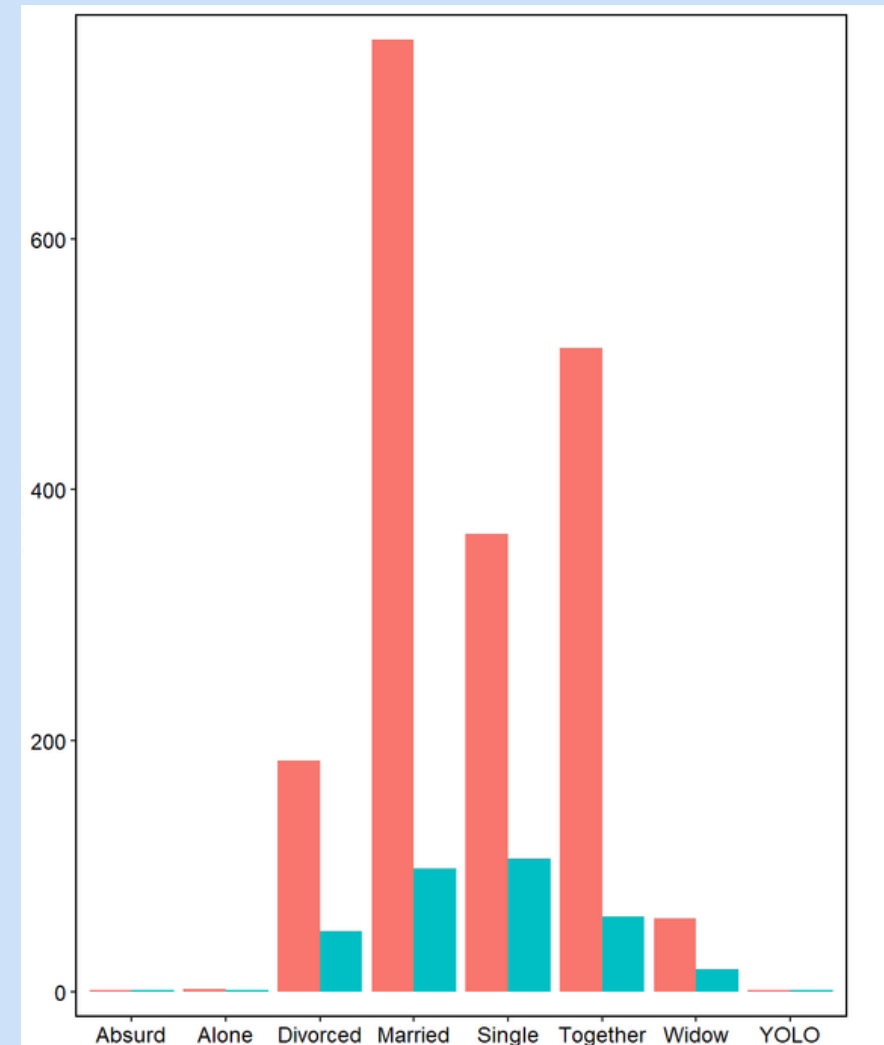
P-value = 1.03×10^{-3}

P-value = 3.53×10^{-13}

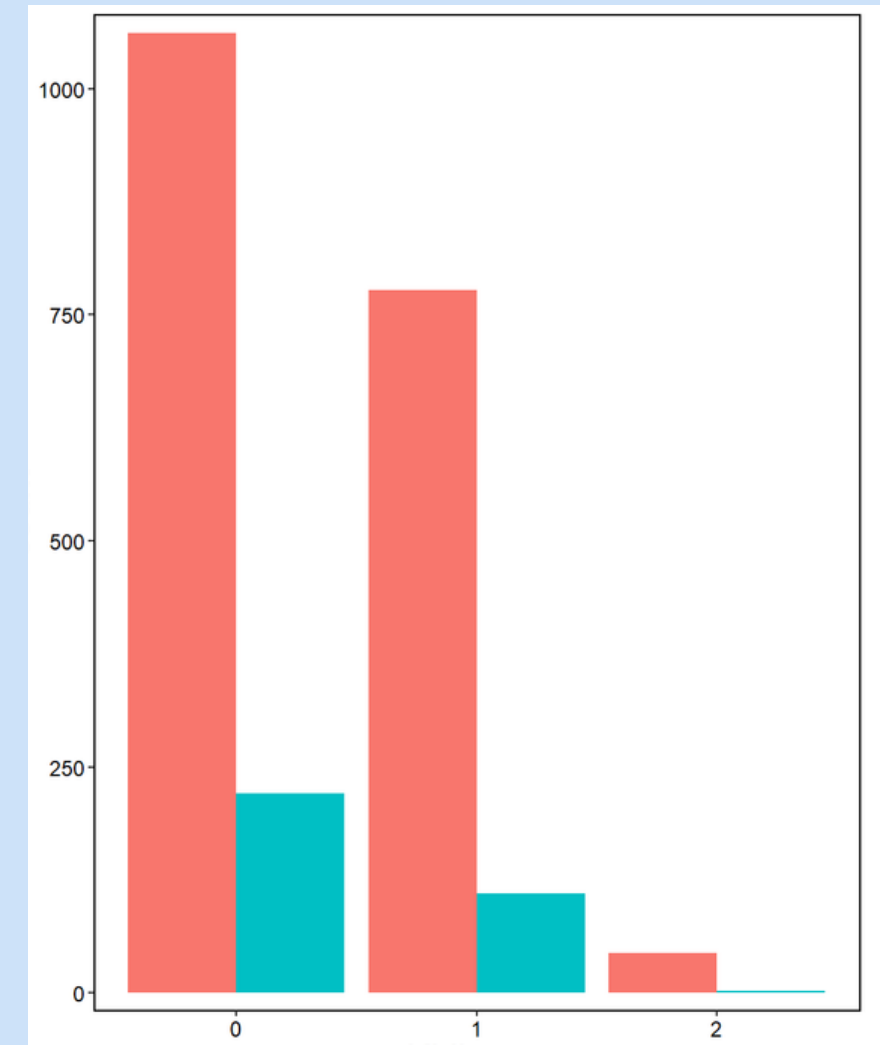
The Number of People



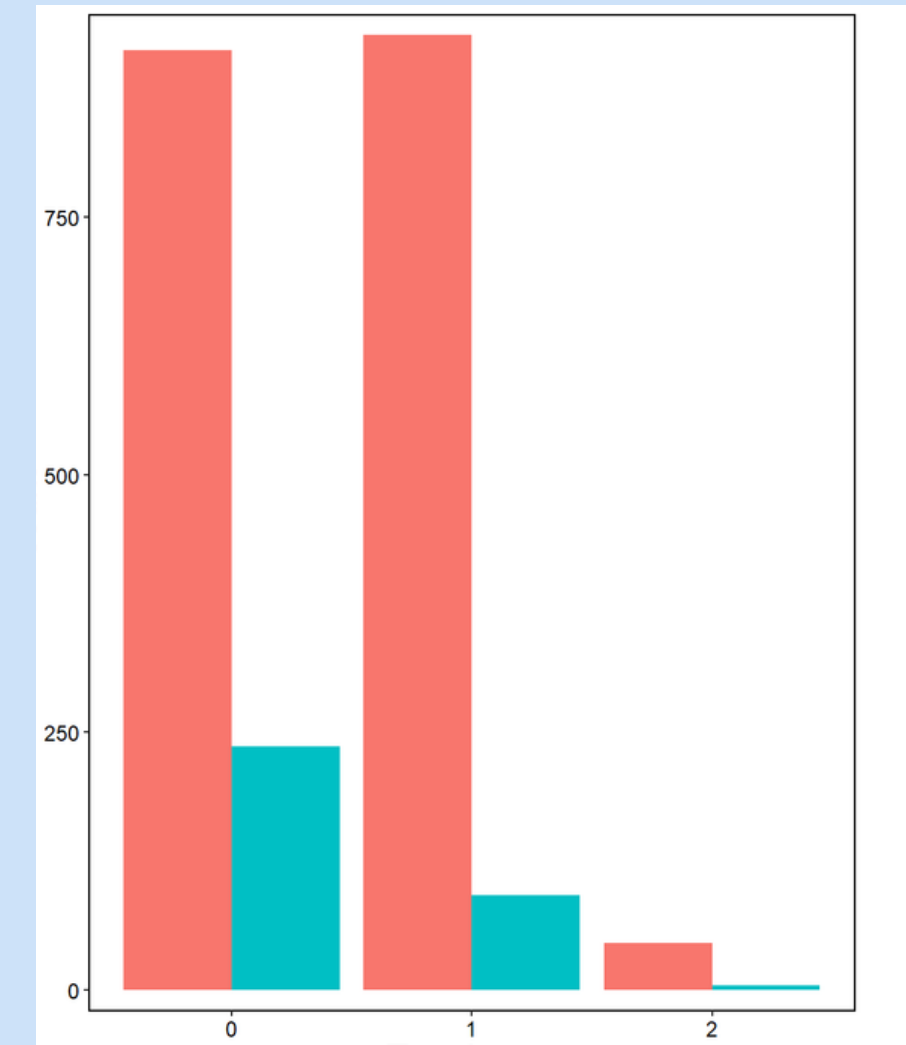
Education



Marital Status



Kidhome



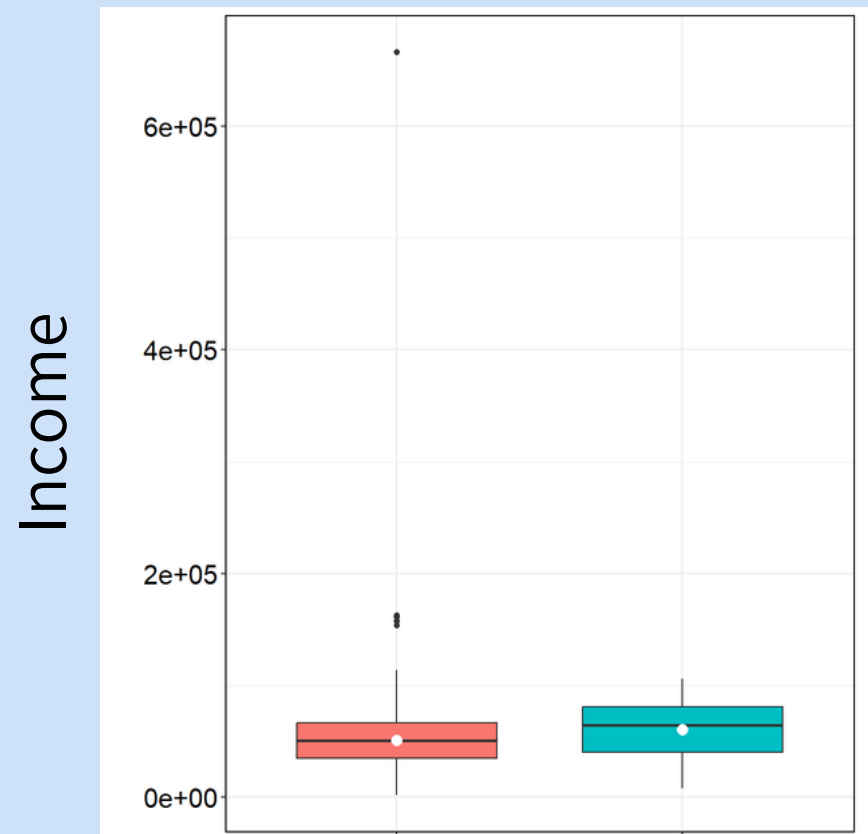
Teenhome

From the Chi-square test, we can say that these factors are related to the purchase by a certain campaign.

Welch Two Sample T-test

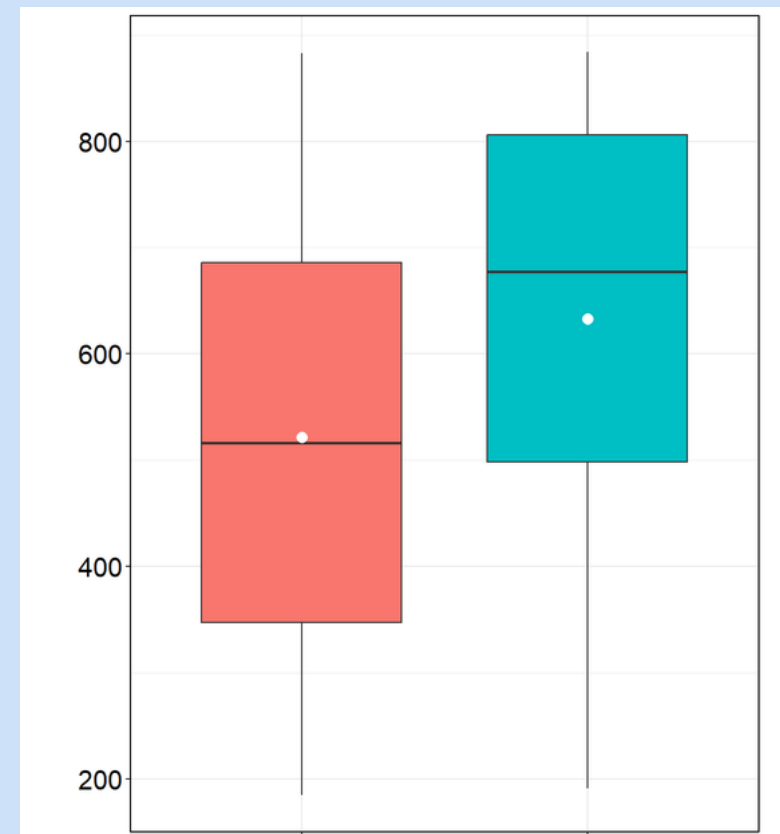
No Buy

Buy



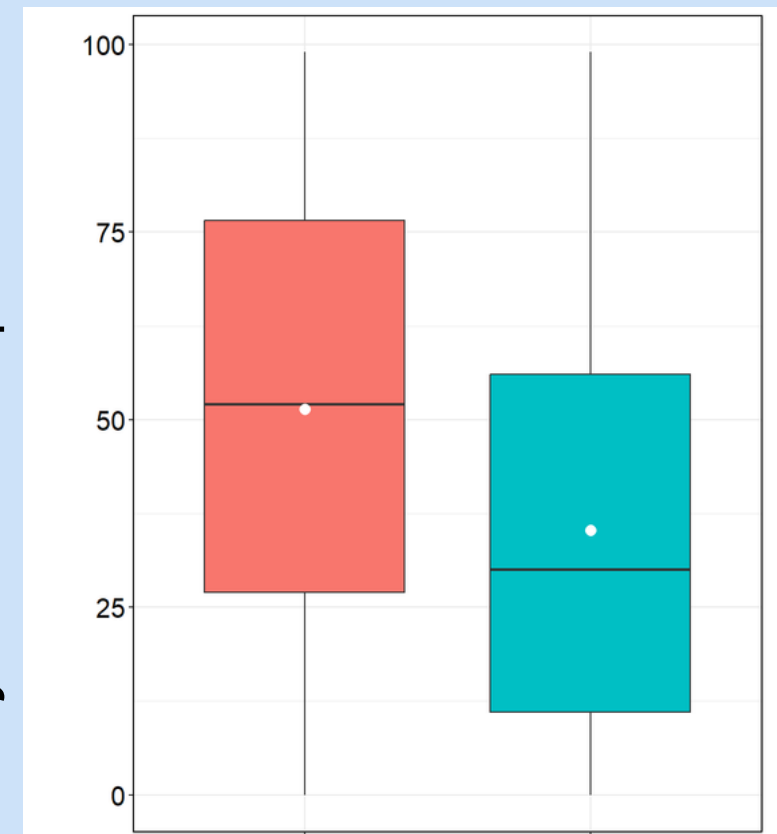
P-value = 4.52×10^{-13}

Number of days elapsed
since admission



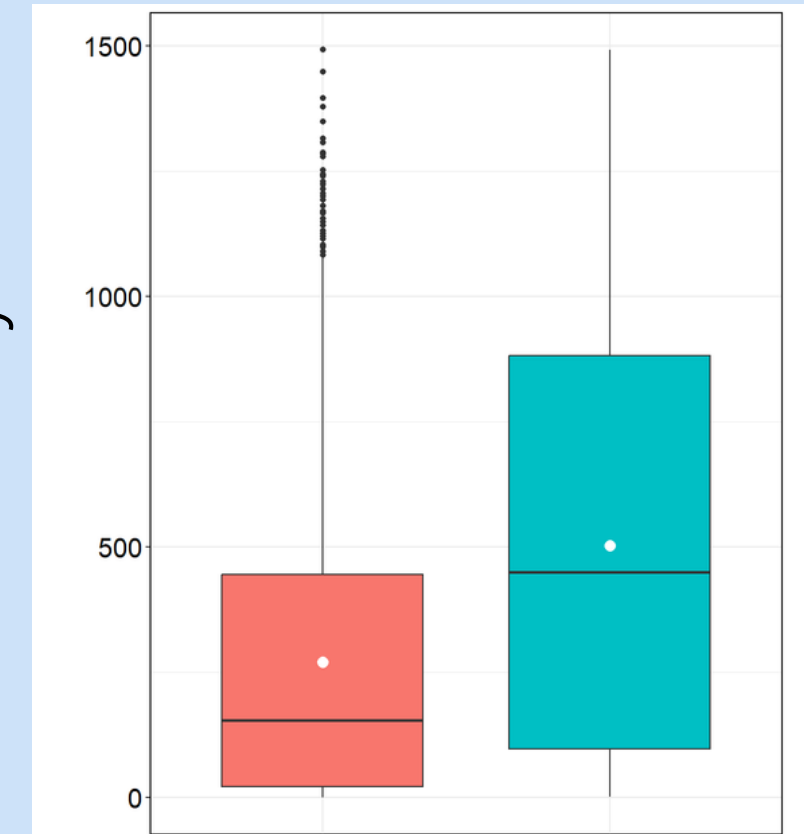
P-value = 2.20×10^{-16}

Days since last purchase



P-value = 2.20×10^{-16}

Amount spent on Wine products
in the last 2 years

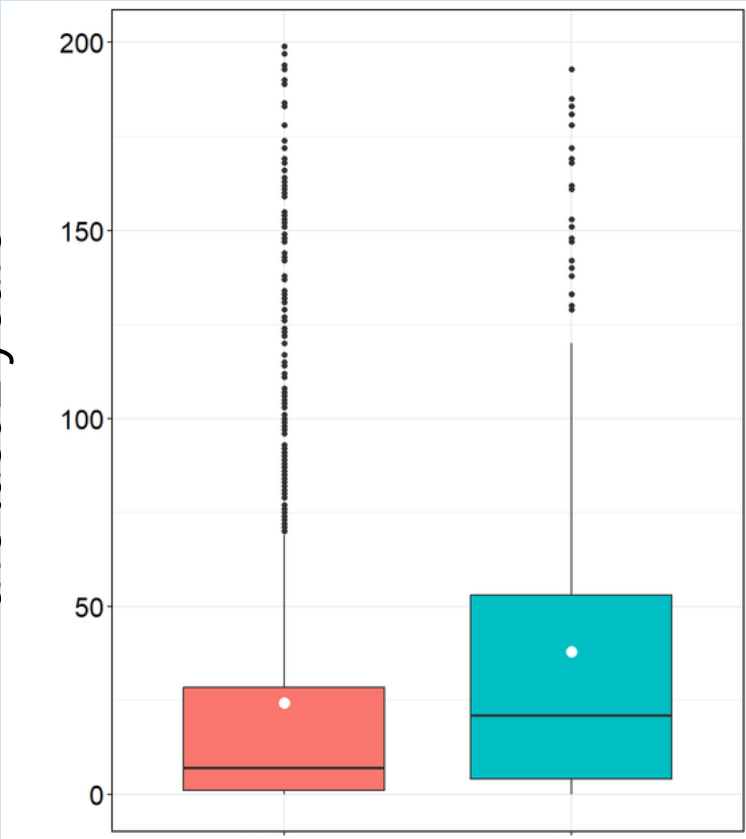


P-value = 4.52×10^{-13}

No Buy

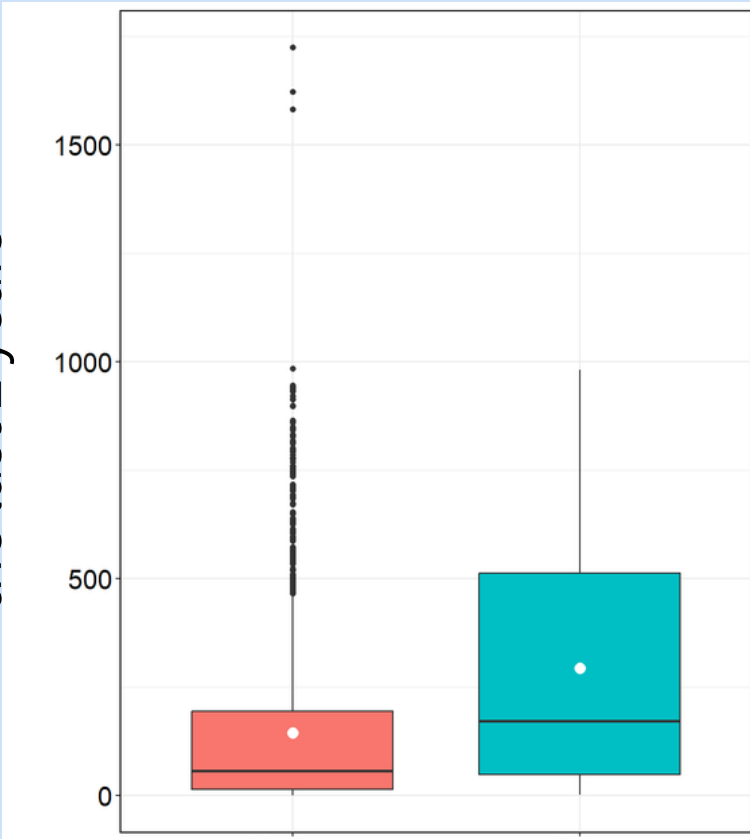
Buy

Amount spent on Fruit products in
the last 2 years



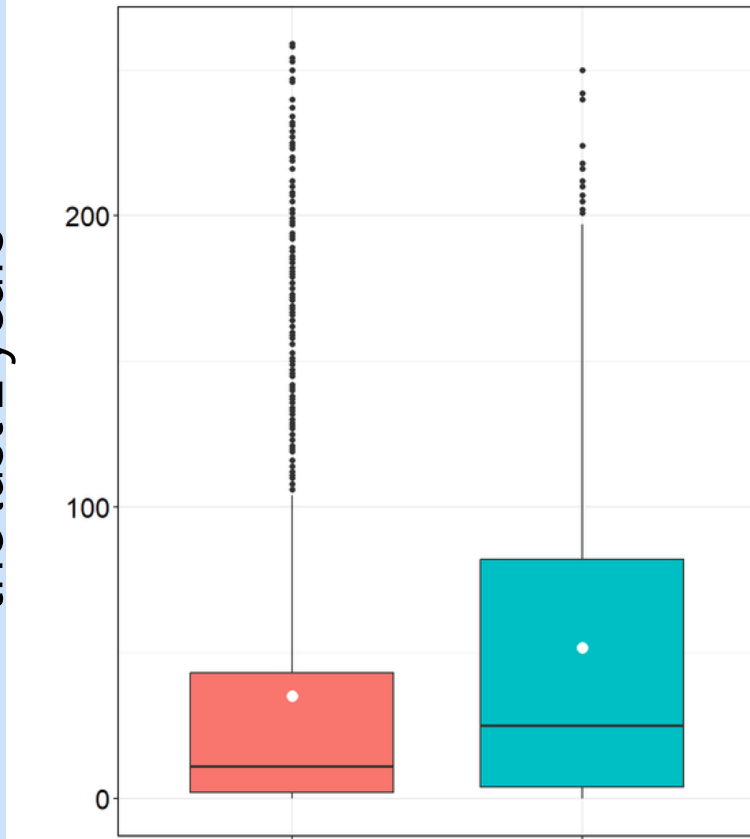
P-value = 2.46×10^{-10}

Amount spent on Meat products in
the last 2 years



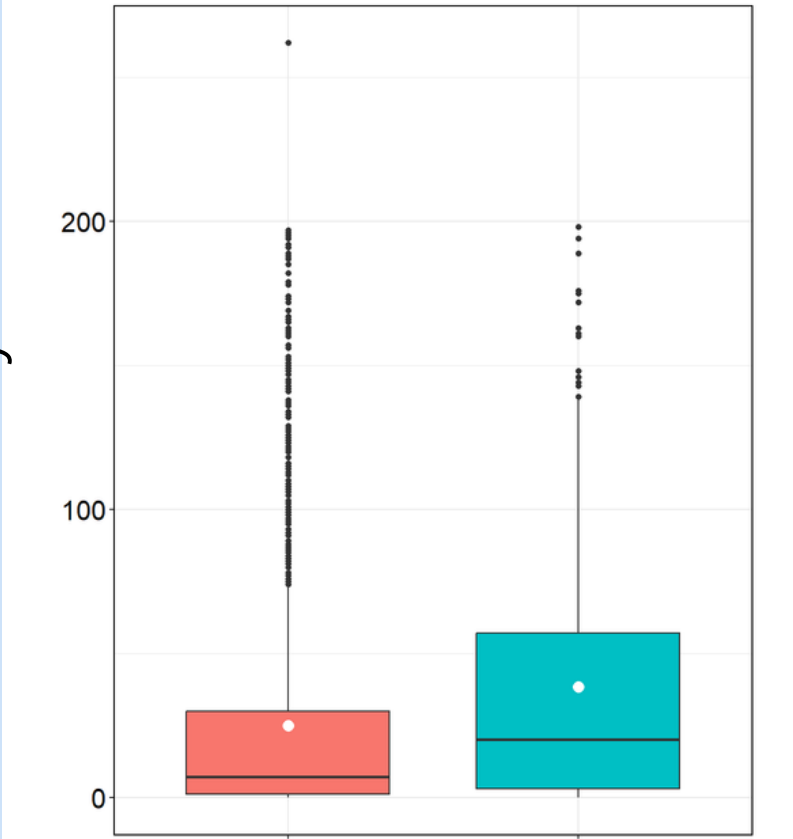
P-value = 2.20×10^{-16}

Amount spent on Fish products in
the last 2 years



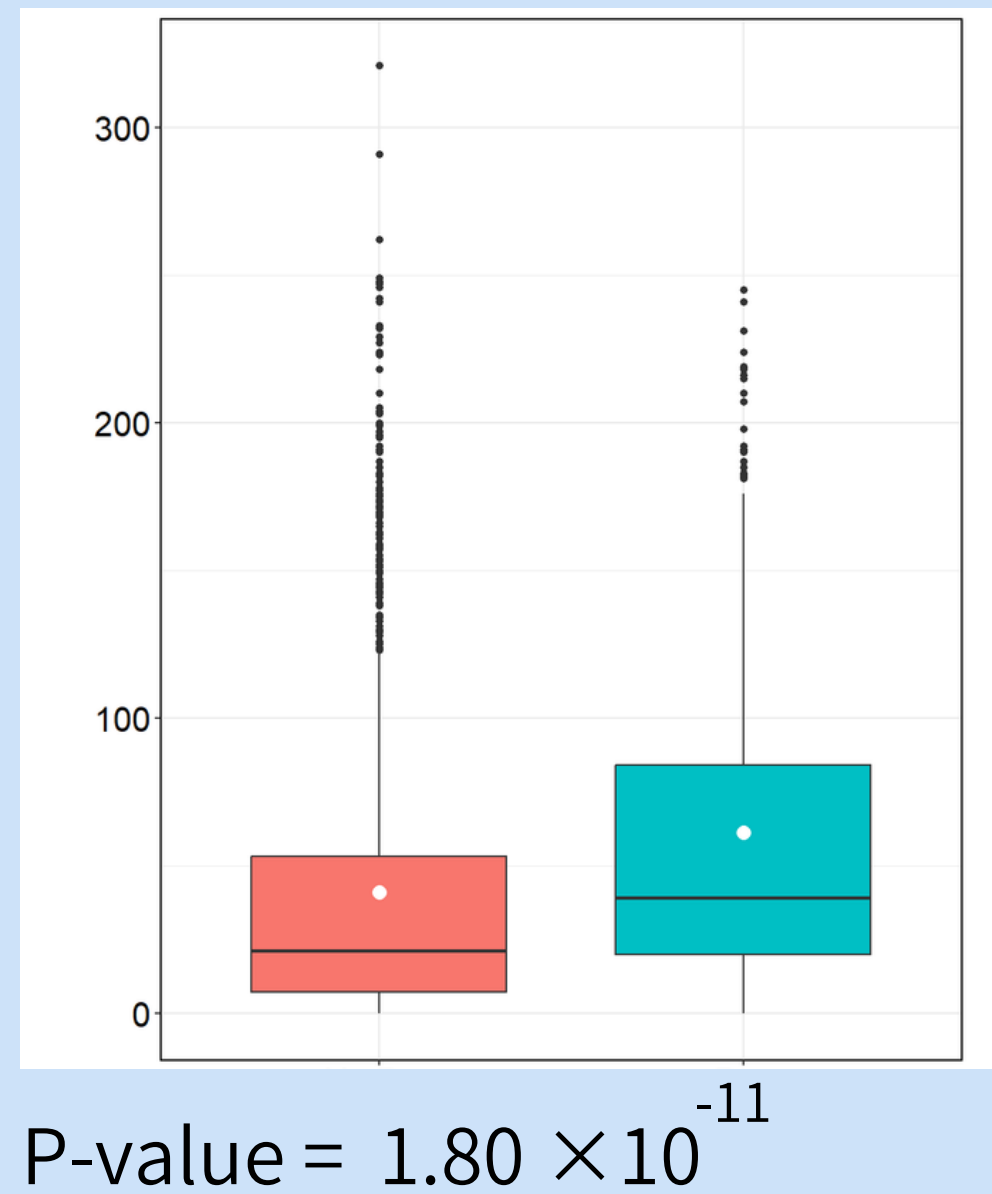
P-value = 8.83×10^{-7}

Amount spent on Sweet products in
the last 2 years

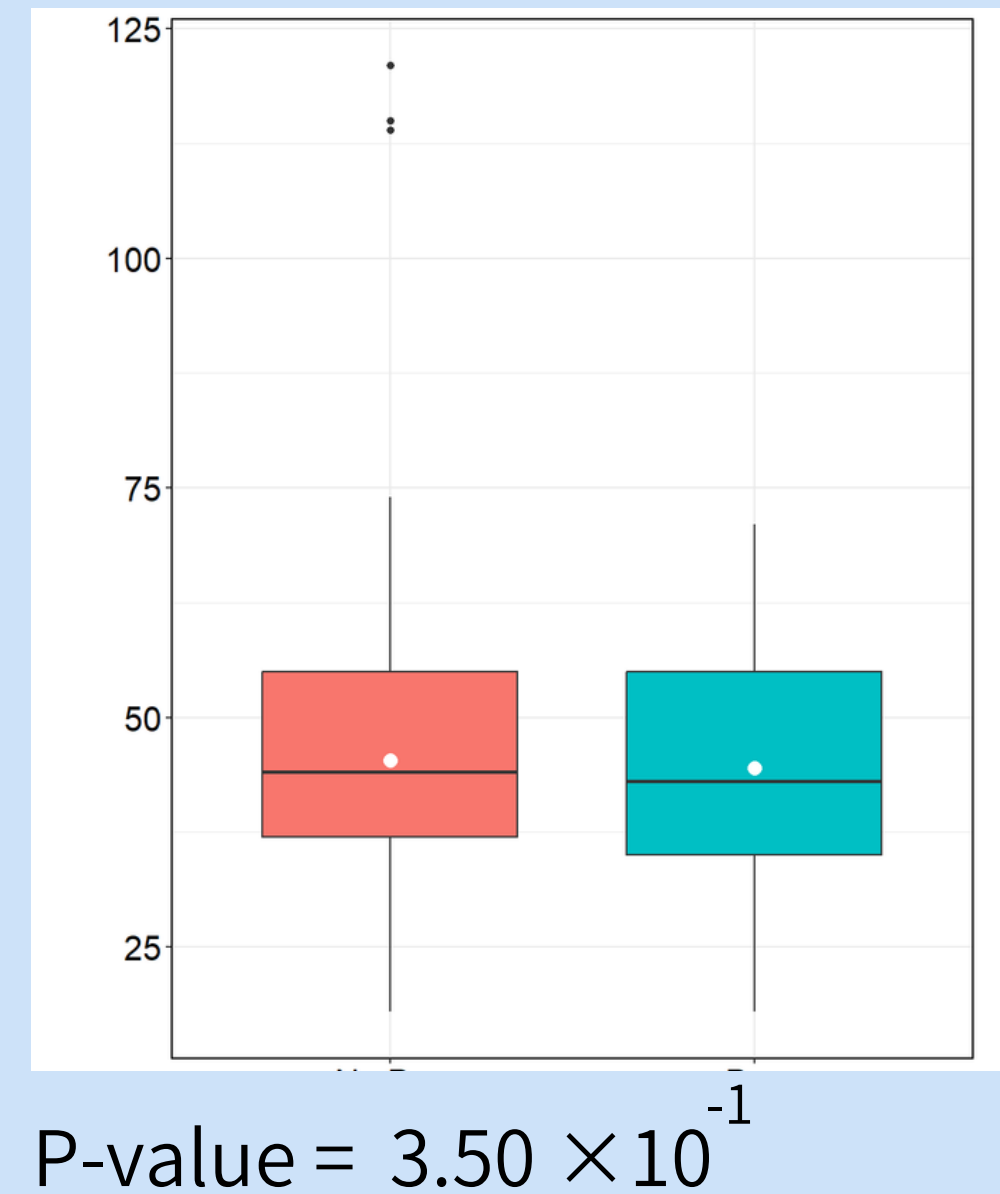


P-value = 1.60×10^{-9}

Amount spent on Gold products
in the last 2 years



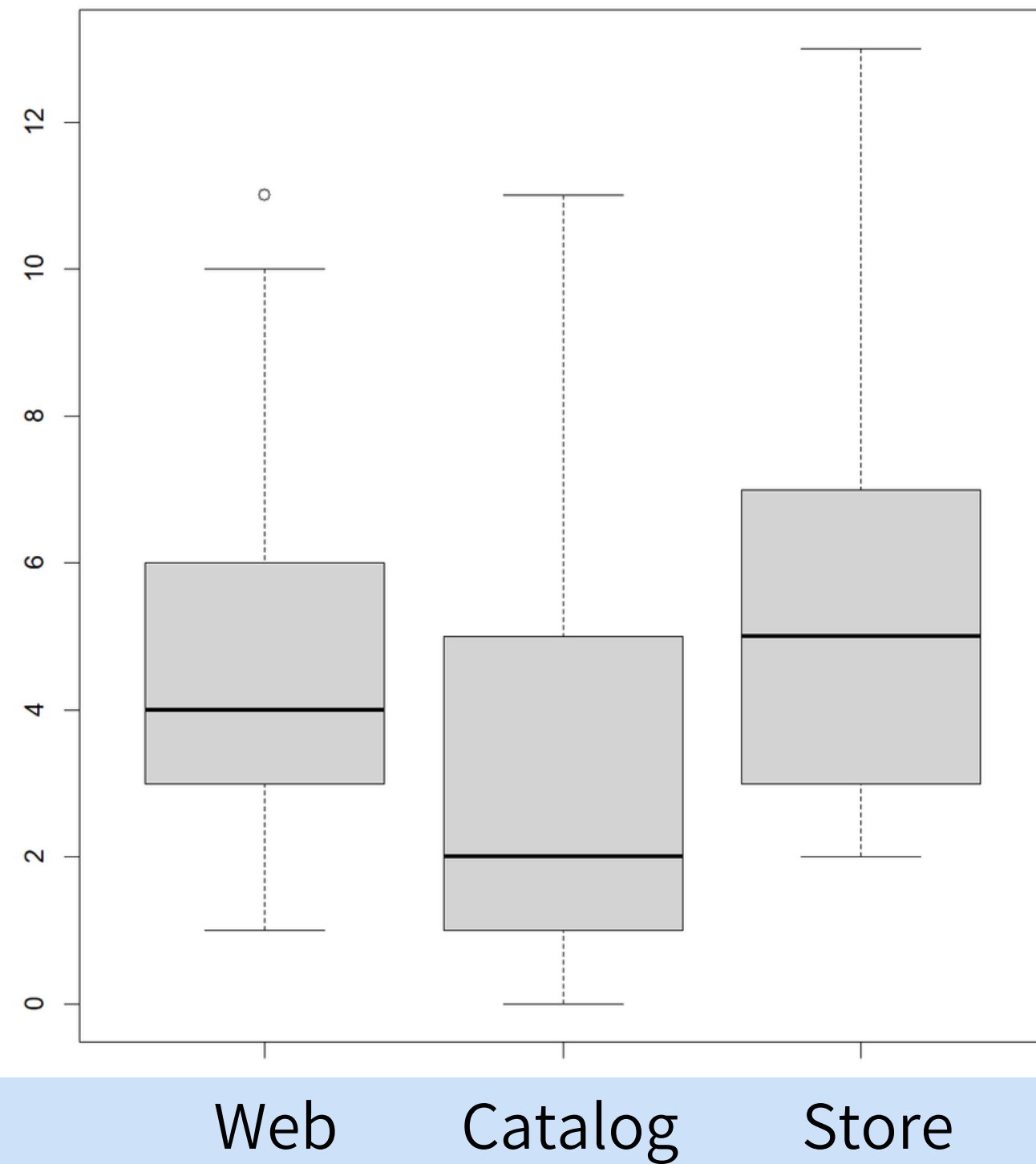
Age



No Buy
Buy

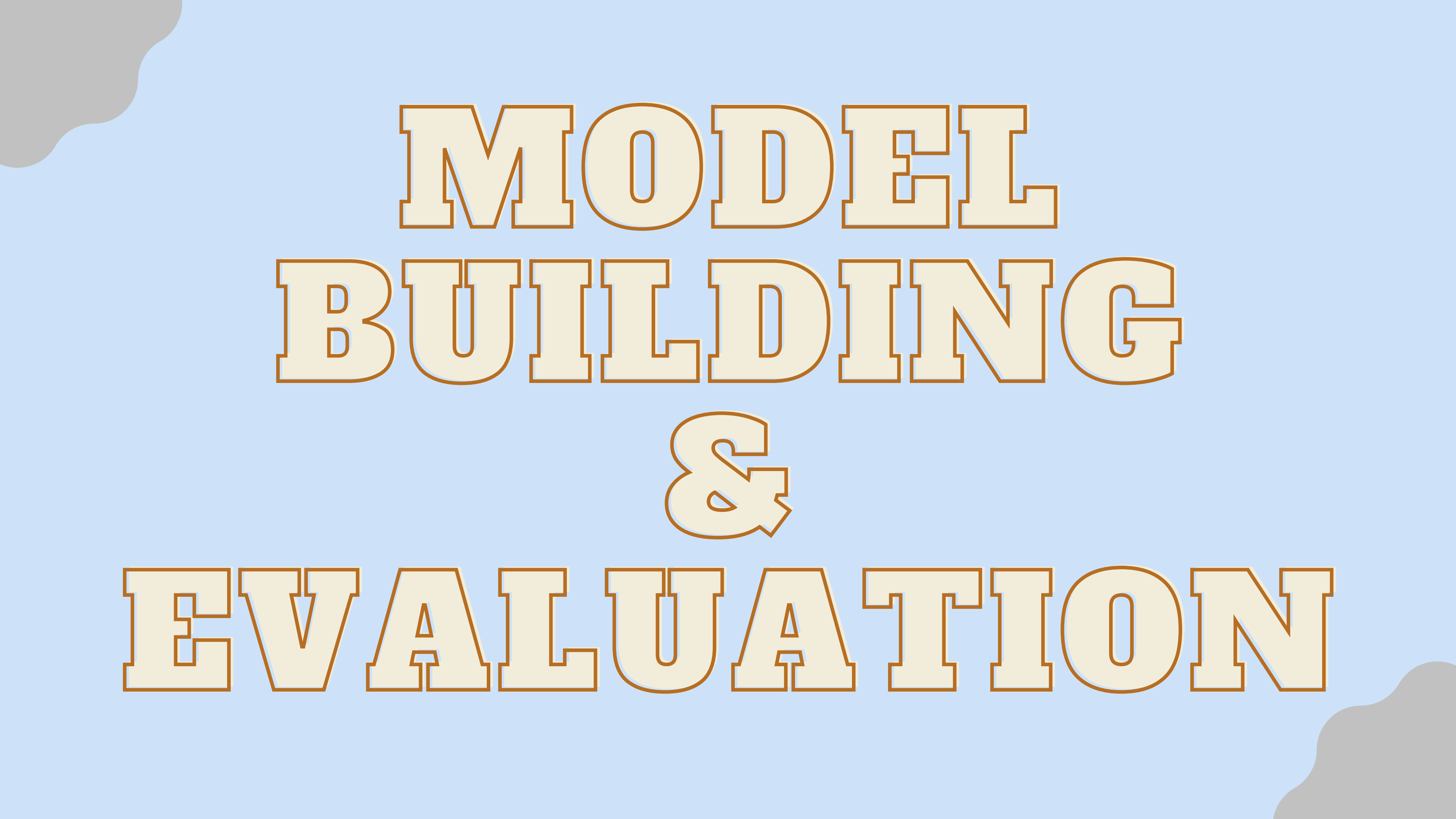
- Tests for factors other than age showed significant differences between the groups that purchased and those that did not
- Customers who have been members for a long time and purchase more frequently are purchasing more products in this campaign

Number of purchases made
by campaign buyers



Characteristics of Buyers from this Campaign

- Increased purchases of all products
- Long-term members and more frequent buyers than non-buyers
- Single, childless people tend to buy more
- Stores have the highest average number of purchases



MODEL BUILDING & EVALUATION

The summary of Logistic Regression model

Logistic Regression with all features

Evaluation Metric

PR AUC (CV): 0.500



Build a better model
with reference to P-values and EDA

Logit Regression Results						
Dep. Variable:	Response	No. Observations:	2216			
Model:	Logit	Df Residuals:	2186			
Method:	MLE	Df Model:	29			
Date:	Sun, 09 Jul 2023	Pseudo R-squ.:	0.2860			
Time:	01:59:51	Log-Likelihood:	-669.52			
converged:	True	LL-Null:	-937.76			
Covariance Type:	nonrobust	LLR p-value:	8.843e-95			
	coef	std err	z	P> z	[0.025	0.975]
const	-1.3109	4.99e+06	-2.63e-07	1.000	-9.78e+06	9.78e+06
Income	0.0568	0.075	0.760	0.447	-0.090	0.203
Kidhome	0.3382	0.195	1.734	0.083	-0.044	0.720
Teenhome	-1.1075	0.185	-5.996	0.000	-1.469	-0.745
Recency	-0.8162	0.078	-10.447	0.000	-0.969	-0.663
MntWines	0.5508	0.098	5.649	0.000	0.360	0.742
MntFruits	0.0174	0.084	0.208	0.835	-0.146	0.181
MntMeatProducts	0.3559	0.101	3.519	0.000	0.158	0.554
MntFishProducts	-0.1473	0.092	-1.598	0.110	-0.328	0.033
MntSweetProducts	0.0818	0.085	0.966	0.334	-0.084	0.248
MntGoldProds	0.1915	0.075	2.539	0.011	0.044	0.339
NumDealsPurchases	0.0430	0.085	0.504	0.614	-0.124	0.210
NumWebPurchases	0.2015	0.084	2.413	0.016	0.038	0.365
NumCatalogPurchases	0.3124	0.107	2.917	0.004	0.102	0.522
NumStorePurchases	-0.6446	0.101	-6.364	0.000	-0.843	-0.446
NumWebVisitsMonth	0.2710	0.108	2.510	0.012	0.059	0.483
Complain	-0.0008	0.079	-0.010	0.992	-0.157	0.155
Age	-0.0134	0.076	-0.176	0.861	-0.163	0.137
Days	0.5728	0.083	6.918	0.000	0.411	0.735
Education_2n Cycle	-0.2671	4.35e+06	-6.13e-08	1.000	-8.53e+06	8.53e+06
Education_Basic	-1.5789	4.35e+06	-3.63e-07	1.000	-8.53e+06	8.53e+06
Education_Graduation	-0.0868	4.35e+06	-1.99e-08	1.000	-8.53e+06	8.53e+06
Education_Master	0.0773	4.35e+06	1.77e-08	1.000	-8.53e+06	8.53e+06
Education_PhD	0.5446	4.35e+06	1.25e-07	1.000	-8.53e+06	8.53e+06
Marital_Status_Absurd	0.9620	5.65e+06	1.7e-07	1.000	-1.11e+07	1.11e+07
Marital_Status_Alone	0.7811	5.65e+06	1.38e-07	1.000	-1.11e+07	1.11e+07
Marital_Status_Divorced	-0.1014	5.65e+06	-1.8e-08	1.000	-1.11e+07	1.11e+07
Marital_Status_Married	-1.1947	5.65e+06	-2.12e-07	1.000	-1.11e+07	1.11e+07
Marital_Status_Single	-0.1368	5.65e+06	-2.42e-08	1.000	-1.11e+07	1.11e+07
Marital_Status_Together	-1.2036	5.65e+06	-2.13e-07	1.000	-1.11e+07	1.11e+07
Marital_Status_Widow	-0.0232	5.65e+06	-4.11e-09	1.000	-1.11e+07	1.11e+07
Marital_Status_YOLO	-0.3943	5.65e+06	-6.98e-08	1.000	-1.11e+07	1.11e+07

Categorical
values

Logistic Regression model with feature engineering

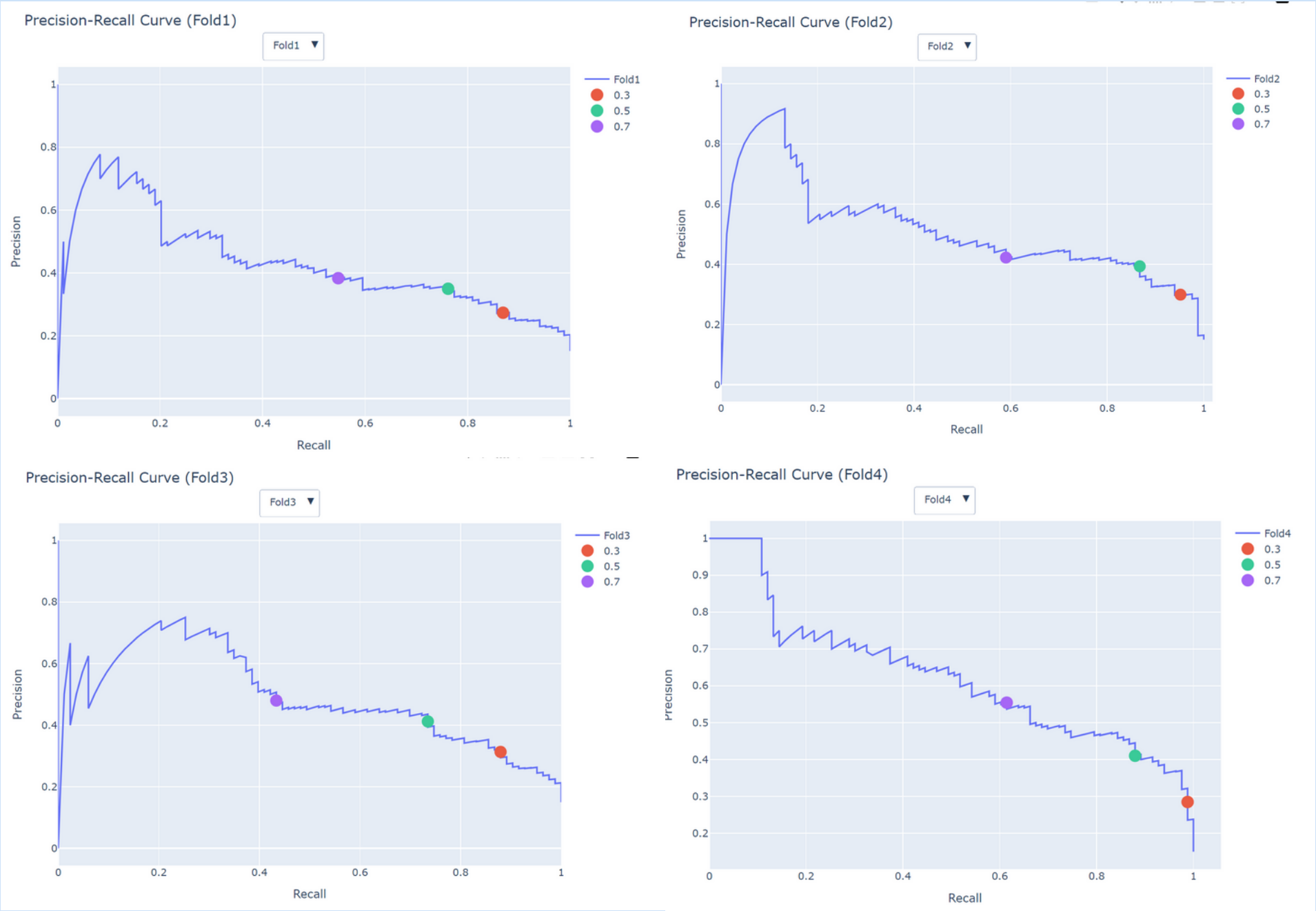
Evaluation Metric
PR AUC (CV): 0.513



The summary of new Logistic Regression model

Logit Regression Results						
Dep. Variable:	Response	No. Observations:	2216			
Model:	Logit	Df Residuals:	2193			
Method:	MLE	Df Model:	22			
Date:	Sun, 09 Jul 2023	Pseudo R-squ.:	0.2815			
Time:	02:05:51	Log-Likelihood:	-673.76			
converged:	True	LL-Null:	-937.76			
Covariance Type:	nonrobust	LLR p-value:	1.041e-97			
	coef	std err	z	P> z	[0.025	0.975]
const	-1.2748	2.73e+06	-4.67e-07	1.000	-5.35e+06	5.35e+06
Kidhome	0.3263	0.183	1.782	0.075	-0.033	0.685
Teenhome	-1.0631	0.161	-6.583	0.000	-1.380	-0.747
Recency	-0.8141	0.078	-10.458	0.000	-0.967	-0.662
MntWines	0.5615	0.096	5.834	0.000	0.373	0.750
MntMeatProducts	0.3821	0.098	3.889	0.000	0.190	0.575
MntFishProducts	-0.0887	0.086	-1.031	0.302	-0.257	0.080
NumWebPurchases	0.2583	0.081	3.190	0.001	0.100	0.417
NumCatalogPurchases	0.3658	0.105	3.476	0.001	0.160	0.572
NumStorePurchases	-0.6204	0.100	-6.216	0.000	-0.816	-0.425
NumWebVisitsMonth	0.2501	0.100	2.502	0.012	0.054	0.446
Days	0.5872	0.082	7.162	0.000	0.427	0.748
Education_2n Cycle	-0.2229	2.3e+07	-9.7e-09	1.000	-4.5e+07	4.5e+07
Education_Basic	-1.5558	2.3e+07	-6.77e-08	1.000	-4.5e+07	4.5e+07
Education_Graduation	-0.0425	2.3e+07	-1.85e-09	1.000	-4.5e+07	4.5e+07
Education_Master	0.0654	2.3e+07	2.84e-09	1.000	-4.5e+07	4.5e+07
Education_PhD	0.4811	2.3e+07	2.09e-08	1.000	-4.5e+07	4.5e+07
Marital_Status_Absurd	1.2655	2.89e+07	4.39e-08	1.000	-5.65e+07	5.65e+07
Marital_Status_Alone	0.7342	2.89e+07	2.54e-08	1.000	-5.65e+07	5.65e+07
Marital_Status_Divorced	-0.1364	2.89e+07	-4.73e-09	1.000	-5.65e+07	5.65e+07
Marital_Status_Married	-1.2303	2.89e+07	-4.26e-08	1.000	-5.65e+07	5.65e+07
Marital_Status_Single	-0.1957	2.89e+07	-6.78e-09	1.000	-5.65e+07	5.65e+07
Marital_Status_Together	-1.2497	2.89e+07	-4.33e-08	1.000	-5.65e+07	5.65e+07
Marital_Status_Widow	-0.0668	2.89e+07	-2.32e-09	1.000	-5.65e+07	5.65e+07
Marital_Status_YOLO	-0.3955	2.89e+07	-1.37e-08	1.000	-5.65e+07	5.65e+07

Precision-Recall Curve and Threshold points for each fold



Precision: the proportion of the data points our model says existed in the relevant class that were indeed relevant

Recall: the ability to find all relevant instances of a class in a dataset 🌸

Considering the trade-off between Precision and Recall, it would be good to set the threshold value to 0.5

CONCLUSION

1. The most important element is recency, which is that when customers' the last date of purchase is closer, they are more likely to respond.
2. The longer date of membership is, they are more likely to respond.
3. The fewer teenagers in the household, they are more likely to respond.

CITATION

1. investopedia, <https://www.investopedia.com/terms/m/marketing.asp>
2. <https://emeritus.org/in/learn/what-is-the-importance-of-marketing-for-business/>
3. lumenlearning, <https://courses.lumenlearning.com/suny-wmopen-introbusiness/chapter/the-role-of-customers-in-marketing/#:~:text=Marketing%20identifies%20customers%2C%20their%20needs,proportional%20to%20what%20it%20costs.>



**THANK YOU
FOR YOUR
LISTENING**